

unit's digit of its terms, so unit's digit of each one of $N_1, N_2, \dots, N_{10}$ is also the same.

$\therefore$ Unit's digit in $N = 10 \times$ Unit's digit of $N_1 = 0$.

- 353.** Clearly, each of the 38 terms $\left(\frac{1}{4}\right), \left(\frac{1}{4} + \frac{1}{50}\right), \left(\frac{1}{4} + \frac{2}{50}\right), \dots, \left(\frac{1}{4} + \frac{37}{50}\right)$ has a value lying between 0 and 1, while each one of the 12 terms $\left(\frac{1}{4} + \frac{38}{50}\right), \left(\frac{1}{4} + \frac{39}{50}\right), \dots, \left(\frac{1}{4} + \frac{49}{50}\right)$ has a value lying between 1 and 2.

Hence, the given expression $= (0 \times 38) + (1 \times 12) = 12$.

- 354.** $100^{25} - 25 = (10^2)^{25} - 25 = 10^{50} - 25$
 $= \underbrace{1000 \dots 00}_{50 \text{ zeros}} - 25 = \underbrace{9999 \dots 9975}_{48 \text{ times}}$
 $\therefore$ Sum of digits $= (48 \times 9) + 7 + 5 = 432 + 7 + 5 = 444$.

- 355.** $(1024)^4 \times (125)^{11} = (2^{10})^4 \times (5^3)^{11} = 2^{40} \times 5^{33} = 2^7 \times (2^{33} \times 5^{33}) = 2^7 \times 10^{33} = 128 \times 10^{33}$.

Clearly, the number has 1, 2, 8 and thirty-three zeros, i.e. $(3 + 33) = 36$ digits in all.

- 356.** From 300 to 399, we note that when '4' comes only one time = 19 such instances.

From 400 to 499, we note that when '4' comes only one time = 80 such instances.

So, total $= (19 + 80) = 99$ such instances

- 357.** $21 = 3 \times 7$ is not a prime number because 21 is a composite number.

- 358.** Given $x = a(b - c)$, $y = b(c - a)$; $z = (a - b)$

$$x = a(b - c)$$

$$\Rightarrow \frac{x}{a} = b - c \dots (i)$$

Similarly, $y = b(c - a)$

$$\Rightarrow \frac{y}{b} = c - a \text{ (ii) and similarly } z = c(a - b) \frac{z}{c} = c - a \text{ (iii)}$$

Adding (i), (ii) and (iii) we get

$$\therefore \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = b - c + c - a + a - b = 0$$

$$\Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 0$$

$$\therefore \left(\frac{x}{a}\right)^3 + \left(\frac{y}{b}\right)^3 + \left(\frac{z}{c}\right)^3$$

$$= 3 \times \frac{x}{a} \times \frac{y}{b} \times \frac{z}{c} = \frac{3xyz}{abc}$$

$$[\text{If } a + b + c = 0, a^3 + b^3 + c^3 = 3abc]$$

- 359.** Every real number is a rational number is not a correct statement.

- 360.** Given

$$a + b + c = 6$$

$$ab + bc + ca = 10$$

$$\therefore (a + b + c)^2 = 36$$

$$\Rightarrow a^2 + b^2 + c^2 + 2ab + 2bc + 2ca = 36$$

$$\Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) = 36$$

$$\Rightarrow a^2 + b^2 + c^2 + 2 \times 10 = 36$$

$$\Rightarrow a^2 + b^2 + c^2 = 16$$

$$\text{As we know } \frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca} = (a + b + c)$$

$$\frac{a^3 + b^3 + c^3 - 3abc}{16 - (ab + bc + ca)} = 6$$

$$\Rightarrow \frac{a^3 + b^3 + c^3 - 3abc}{16 - 10} = 6$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = 6 \times 6$$

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = 36$$

- 361.** Let the blank space is x

$$\therefore 1001 \times 111 = 110000 + 11x$$

Now find the value of x

$$(1000 + 1) \times 111 = 110000 + 11x$$

$$111000 + 111 = 110000 + 11x$$

$$111111 = 110000 + 11x$$

$$11x = 111111 - 110000$$

$$11x = 1111$$

$$x = \frac{1111}{11} = 101$$

- 362.** The place value of 5 in 15201 = 5000

Place value of 6 in 2659 = 600

$$\therefore \text{ we have } 5000 + 600 = 7x$$

$$7x = 5600$$

$$x = \frac{5600}{7} = 800$$

- 363.** Let the two - digit number be $10a + b$ where $a > b$

According to the question,

$$a + b = 12 \dots (i)$$

$$a - b = 6 \dots (ii)$$

On adding equation (i) and (ii).

$$2a = 19$$

$$\Rightarrow a = 9$$

From equation (i),

$$9 + b = 12$$

$$\Rightarrow b = 12 - 9 = 3$$

$$\therefore \text{ Number is } 10a + b$$

$$= 9 \times 10 + 3 = 93$$

$$\therefore \text{ When } a < b. \text{ Then required number is } = 39$$

- 364.** Greatest four digit number that begins with 3 and ends with 5 = 3995

Least four digit number that begins with 3 and ends with 5 = 3005